

Problem

A spectrum analyzer indicates that a signal is made up of three components only: 640 kHz at 2 V, 644 kHz at 1 V, 636 kHz at 1 V. If the signal is applied across a 10-Ω resistor, what is the average power absorbed by the resistor?

Step-by-step solution

Step 1 of 1

$$\begin{aligned} p &= \frac{V_n^2}{R} + \frac{1}{2} \sum \frac{V_n^2}{R} \\ &= 0 + 0.5 \left\{ \frac{[2^2 + 1^2 + 1^2]}{10} \right\} = 300 \text{ mW} \end{aligned}$$